

SNF-Bench: Separating Static Drift from Natural Flow in Long-Horizon Fixed-Camera Video Generation

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Abstract

Long-horizon video generation is evaluated with whole-frame metrics that reward motion and temporal consistency. For fixed-camera nature scenes this creates an ambiguity: motion of water, fire, smoke, or rain is desirable, whereas motion of the background is an error. A system can therefore score well on motion while its scene drifts, or on consistency while its flow stagnates. We introduce SNF-Bench, an evaluation framework for long-horizon fixed-camera generation that partitions each scene into static support and dynamic flow and reports static fidelity, flow persistence with absolute magnitude, and drift leakage separately, never as one score. Drift leakage is interpretive context rather than a headline measurement. Each factor is validated mechanistically rather than by correlation with preference: we inject global translation, rotation, and scale drift and progressive motion attenuation at known severity into real generations, and require each factor to respond in its stated direction and to remain selective against corruptions it does not target. Auditing publicly released long-horizon text-conditioned checkpoints under one recorded common inference configuration, plus an image-conditioned track with released-pipeline references and a deployment-sensitivity panel, we find that whole-frame motion and static-region drift induce near-opposite orderings of the same outputs. At maximum controlled translation, fBD and NBF rise to $1.86\times$ and $1.32\times$ baseline, but whole-frame Dynamic Degree reaches only $1.07\times$ —rewarding the corruption. SNF-Bench measures where motion occurs and whether it persists; it does not measure physical realism.

1. Introduction

Long-horizon, autoregressive (AR), and streaming video diffusion models now extend generation from sec-

onds to minute-scale horizons [20, 32, 34, 36, 38, 39]. Evaluation has not kept pace: small temporal errors invisible in short clips accumulate over long rollouts into background displacement, color drift, or motion decay, and whole-frame metrics summarize these qualitatively different failures into one score.

The problem is sharpest in fixed-camera nature scenes. In a static-camera video of a river, the water should move while the bank, rocks, and surrounding landscape stay fixed. The same separation governs waterfalls, ocean waves, rainfall, fire, smoke, and wind-driven vegetation: some regions form static support that must remain structurally fixed, while others carry dynamic flow that must persist. Whole-frame motion measures reward optical flow regardless of its origin, crediting background drift as readily as genuine fluid motion, while temporal-consistency measures can favor videos whose stability comes from suppressing the flow they should sustain. Two opposite failures—static drift and flow decay—are thereby conflated, and increasingly so at long horizons, where accumulated background motion can dominate a sequence’s apparent dynamics. Figure 1 shows both failures in outputs of released checkpoints under the common setting.

We introduce SNF-Bench (Static-Natural Flow), an evaluation framework built around this decomposition, evaluated at 60 and 120s horizons with shorter (5s) and longer (240s) tiers retained as diagnostics. Measurement is organized into three factor groups, the third interpretive. Static fidelity captures geometric displacement and residual flow inside nominally stationary regions. Flow persistence reports the retention of dynamic-region motion with its absolute compensated magnitude, so a ratio cannot look strong merely because motion is uniformly weak. Drift leakage relates static-region motion and global-motion compensation to dynamic-region flow; it contextualizes the two measurement axes rather than adding a third, and the central claim does not rest on it.

A benchmark is only as trustworthy as its measure-

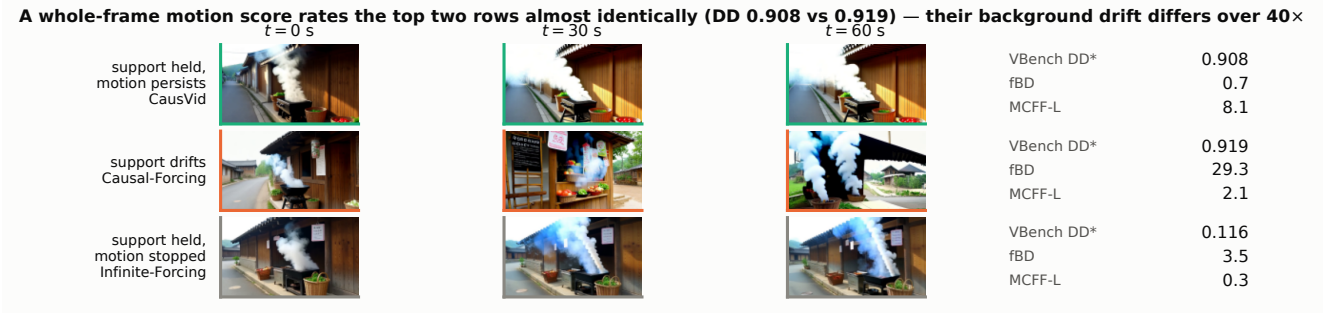


Figure 1. Why a fixed-camera benchmark needs a spatial decomposition. Three 60s outputs from released checkpoints under the recorded common configuration. A whole-frame motion score rates the first two rows almost identically, yet only the first holds its scene in place; their static-region drift differs by over an order of magnitude. The third row shows the opposite failure, which that score does detect. DD* is the per-video quantity underlying VBench [12] Dynamic Degree, computed under its published rule but read before the binarization the suite applies when aggregating, so that near-identical rows remain distinguishable as numbers.

ments, so we validate the protocol independently of any model comparison. Controlled perturbations inject global translation, rotation, and scale drift, progressive motion attenuation, late freezing, photometric drift, and temporal repetition into real fixed-camera generations at known severity, and we require each factor to respond in the direction its definition states and to remain selective against corruptions it does not target. Because a parameter expressed in degrees is not commensurate with one expressed in pixels, geometric severity is specified as the mean displacement a corruption induces, which makes the families directly comparable. We additionally recompute every measurement under systematic erosion and dilation of the region partition.

We then audit publicly released long-horizon generators as executed through the recorded evaluation pipeline. The audit exposes cases where conventional metrics and SNF-Bench interpret the same outputs differently: high whole-frame motion coexisting with substantial static-region displacement, and high stability coexisting with decaying flow. We do not argue that existing metrics are incorrect. They answer a whole-frame question accurately; that question does not encode the spatial distinction this setting requires. Nor does SNF-Bench measure physical realism—a sequence may be drift-free and persistent yet physically implausible. The contribution is narrower and complementary: making the spatial origin and temporal persistence of motion explicit.

Our contributions are:

1. SNF-Bench, an evaluation framework for long-horizon fixed-camera generation organized around a static-support/dynamic-flow partition.
2. A measurement protocol that reports static fidelity

and flow persistence with absolute magnitude as the primary axes, with drift leakage provided separately as interpretive context; the factors are never collapsed into one score.

3. Mechanistic validation of every reported factor under corruptions of known type and severity, with selectivity measured against the families each factor should ignore and every measurement recomputed under erosion and dilation of the region partition.
4. A public-model audit showing that this decomposition materially changes the interpretation and ordering of long-horizon behavior relative to whole-frame metrics.

2. Related Work

Video-generation evaluation. General-purpose suites assess visual quality, temporal consistency, motion, and semantic alignment. VBench [12] decomposes quality into dimensions including background consistency, motion smoothness and dynamic degree, and VBench-2.0 [41] extends it toward intrinsic faithfulness; EvalCrafter [21] learns a weighting of many metrics from user opinion; Video-Bench [7] scores generations with multimodal language models validated against human preference. Others specialize: T2V-CompBench [28] targets compositional binding, VM-Bench [19] perceptually aligned motion quality, and WorldScore [4] controllability and dynamics for world generation. These set the standards a benchmark paper must meet—explicit taxonomy, validated metrics, stated scale—but their measurements are whole-frame or sequence-level, and such an aggregate cannot reveal whether motion originates where intended or in displacement of nominally static structure. VM-Bench is the closest comparator, but it asks whether

145 motion is perceptually plausible where we ask where
146 it occurs and whether it survives compensation for
147 global displacement. Our validation is correspondingly
148 mechanistic—perturb by a known amount, require
149 the expected response—complementing the human-
150 alignment evidence those suites provide.

151 Long-horizon autoregressive generation. Autoregres-
152 sive and streaming generators [2, 13, 18, 29, 42]
153 extend synthesis beyond short bidirectional clips
154 by producing frames or temporal chunks causally,
155 including CausVid [37], Self-Forcing [11], Infinite-
156 Forcing [3], Rolling-Forcing [20], Reward-Forcing [22],
157 LongLive [34], and Causal-Forcing [43] with its few-
158 step successor Causal-Forcing++ [40]; bidirectional
159 diffusion systems [1, 6, 14, 31, 35] provide short-horizon
160 reference points. These systems differ substantially in
161 temporal block size, training objective, memory and
162 cache policy, and horizon-extension mechanism [36],
163 yet their evaluation typically relies on generic video
164 metrics or method-specific qualitative analysis. That
165 diversity motivates a recorded common-configuration
166 audit, and the evaluation gap motivates a system-
167 agnostic question: does static support remain fixed
168 while dynamic flow persists?

169 Image dynamics and nature-flow animation. Work on
170 animating still images has long modeled natural mo-
171 tion explicitly [10, 15, 17, 23, 33]: controllable fluid
172 animation separates static image structure from speci-
173 fied fluid motion [24], and Generative Image Dynamics
174 models long-term scene motion through learned trajec-
175 tory representations [16]. These works treat static sup-
176 port and dynamic content as distinct modeling targets;
177 SNF-Bench elevates that distinction to an evaluation
178 principle for long-horizon generators.

179 3. SNF-Bench Design

180 SNF-Bench evaluates long-horizon generation under a
181 deliberately constrained setting: a fixed camera observ-
182 ing a scene containing both stationary structure and
183 persistent natural motion. This isolates two failure
184 modes that are difficult to separate in unconstrained
185 video—static drift, where nominally stationary struc-
186 ture moves over time, and flow decay, where intended
187 dynamics weaken or disappear. Every sequence is
188 treated as a composition of two spatial roles, Ω_{static}
189 for regions expected to remain geometrically station-
190 ary and Ω_{flow} for regions expected to exhibit sustained
191 motion. The partition defines an evaluation role, not a
192 semantic ontology: a riverbank is static support while
193 the river surface is flow; buildings are static support

194 in a rainfall scene while the falling rain is flow. SNF-
195 Bench does not require dynamics inside Ω_{flow} to follow
196 any physical model. It measures whether motion stays
197 localized to the appropriate regions and persists over
198 time without mixing static support. Every system re-
199 ceives identical conditioning per item, and no prompt
200 is rewritten for a particular method. Both condition-
201 ing modalities are evaluated under the same decom-
202 position, differing only in how static support is estab-
203 lished: T2V builds it in the opening sequence, while
204 I2V takes it from a shared source image, so divergence
205 there is attributable to the system rather than to a dif-
206 ferently imagined scene. The main paper reports T2V
207 at 60/120 s and the I2V deployment panel at 60 s; all
208 four 5/60/120/240 s tiers are supplementary.

209 3.1. Scene Categories and Horizons

210 Prompts are stratified across representative fixed-
211 camera natural processes—channel water, ocean waves,
212 precipitation, fire and smoke, lava, and wind-borne
213 motion—with the category set fixed before the final
214 sweep. The realized composition is dominated by direc-
215 tional and broad-area water motion with additional at-
216 mospheric and combustion categories, and conclusions
217 should be read at that scope. The strata are not equally
218 populated: channel water accounts for 61% of the prin-
219 cipal prompt set. Reported aggregates are means over
220 prompts, not category macro-averages: three of the
221 five populated categories hold two prompts each, so
222 equal category weighting would buy balance at a cost
223 in variance the sample cannot support. We instead re-
224 compute every system both ways and state which con-
225 clusions survive, with complete per-category counts in
226 the supplementary.

227 Four temporal regimes are used: 5 s as an initializa-
228 tion check, 60 s as the primary reported horizon, 120 s
229 as a supporting tier, and 240 s as a diagnostic extreme.
230 Conclusions rest on the 60 s tier; the 120 s tier is re-
231 ported alongside it as a supporting check, and the 5
232 and 240 s tiers are in the supplementary material. A
233 run that cannot reach a target horizon under its de-
234 clared track and setting is reported as unsupported
235 rather than silently omitted.

236 3.2. Region Partition

237 The partition is obtained automatically and identically
238 for every sequence. Mean optical-flow magnitude over
239 the leading 12% of sampled frame pairs is thresholded
240 by Otsu’s method [25] into a dynamic region and its
241 complement; the static region is then eroded and a
242 4% frame border discarded, which together form an ig-
243 nored transition band so that boundary pixels—where
244 the partition is least certain and flow least reliable—

contribute to neither region. Restricting the estimate to an early window reduces circularity from late accumulated drift: at 60s the partition is fixed from the first 7.1s and is therefore not defined by the late-window failure being measured. It still depends on early checkpoint behavior and on the same flow estimator used downstream. Two limitations bound the later conclusions. The partition is automatic, so a system drifting within the first 12% of its rollout can influence its own partition; we therefore treat partitions as measured quantities and recompute every result under systematic erosion and dilation of the boundary (Sec. 5). And a two-way partition cannot represent layered content: rain and snow transit static support, so those pixels are assigned by flow magnitude alone and static-fidelity values in precipitation scenes carry an unmodeled contribution. Derivation, the sensitivity study, and the annotation protocol we regard as the natural strengthening of this component are in the supplementary material.

3.3. Benchmark Manifest

SNF-Bench is defined by a versioned manifest fixed before the final sweep, recording per item: identifier and category, exact prompt, target horizons, resolution and frame-rate metadata, and provenance. After freezing, items are never removed because a system performs poorly; corrections require a new benchmark version. Measurement coverage, partial records, and fBD abstentions are reported in Supplementary Tables S6–S7 rather than silently excluded. Coverage is counted from valid per-video measurements rather than from the presence of a results file, so a run that terminated early is reported as partial.

4. Drift–Motion Evaluation Protocol

SNF-Bench evaluates fixed-camera generation along three factor groups: Static Fidelity and Flow Persistence are the measurement axes, and Drift Leakage (DLR, DAR) contextualizes them rather than adding a third measurement. No individual score is sufficient: low background motion can be achieved by freezing the entire video, and high foreground motion can be produced by drifting the entire scene. We therefore always report static-region stability jointly with both the relative persistence and absolute magnitude of dynamic-region motion. Let I_t denote frame t , Ω_{static} and Ω_{flow} the region masks of Sec. 3.2, and $\mathbf{u}_t$ the optical flow from I_t to I_{t+1} , estimated with RAFT [30]. Each sequence defines an early window $\mathcal{E}$ and a late window $\mathcal{L}$ of equal duration, $\mathcal{E}$ beginning after a warm-up so that generation transients do not define the motion reference; extents are fixed in the specification.

4.1. Static Fidelity

Feature-aligned Background Drift (fBD). ORB features [27] extracted inside Ω_{static} from the first frame are matched against each late-window frame under robust geometric filtering; fBD ($\downarrow$) is the median matched-point displacement, averaged over the late window and expressed as a percentage of the frame diagonal. It captures accumulated geometric displacement of persistent structure and is intentionally insensitive to fine-scale flicker that moves no structure (formal definition in Supp. S1).

Normalized Background Flow (NBF). Complementing accumulated drift, NBF ($\downarrow$) is the mean optical-flow magnitude inside the static mask, averaged over all frame pairs and normalized by frame width for resolution invariance and by the inter-frame interval Δt for frame-rate invariance, giving units of 10^{-3} frame widths per second (Supp. S1). Here Δt is the interval of the sampled pairs actually used for flow estimation: methods are evaluated at their native frame rates, which differ across the roster, and the evaluation pipeline subsamples each clip to a common temporal rate before measurement. Expressing NBF per unit time rather than per frame makes the quantity independent of both the native rate and the sampling policy, so the measure remains comparable if either changes. NBF is a normalized magnitude, not a ratio: dividing by whole-frame flow would couple background leakage to foreground behavior and inflate the measure for methods whose dynamic motion decays.

4.2. Flow Persistence

Global-motion compensation. Coherent scene displacement is removed before dynamic-region motion is evaluated. A single translation vector is insufficient: under rotation or zoom the static-region flow field is not constant, and its component-wise median is near zero for a centered rotation, so a translation-only estimate would leave exactly the corruptions Sec. 5 injects uncompensated. We therefore estimate a robust global similarity transform $T_t(\mathbf{x}) = s_t R_t \mathbf{x} + \mathbf{b}_t$ from static-region correspondences under RANSAC, and subtract the displacement field it induces from the measured flow pointwise, giving a compensated field $\tilde{\mathbf{u}}_t$ (Supp. S1). A similarity rather than an unrestricted affine model is used because affine shear absorbs legitimate local deformation and over-corrects. Where correspondences are insufficient we fall back to median translation and record the fallback per sequence.

Motion-Compensated Foreground Flow (MCFF). MCFF ($\uparrow$) is the mean compensated flow magnitude

inside Ω_{flow} over a temporal window. We report it on both the early and the late window—MCFF-E and MCFF-L—so a method with uniformly negligible motion cannot appear strong through relative measures alone (Supp. S1).

Flow Persistence (FP). FP ($\uparrow$) is the ratio of late to early MCFF, clipped at a fixed constant to prevent unstable ratios when early motion is small (Supp. S1). FP near one indicates retained motion; substantially smaller values indicate decay; values above one are not automatically better. FP is never reported without its MCFF magnitudes.

4.3. Drift Leakage and Attenuation

Whole-frame metrics cannot determine whether apparent dynamic-region motion arises from intended flow or from displacement of the whole scene; we report two diagnostics, since neither subsumes the other. Let F_{static} denote mean flow magnitude over Ω_{static} , and F_{raw} , F_{comp} the mean raw and compensated magnitude over Ω_{flow} , all in the late window, where drift has accumulated.

Drift Leakage Ratio (DLR). DLR ($\downarrow$), the ratio of static-region to raw dynamic-region flow magnitude in the late window, locates the sequence’s motion energy (Supp. S1). It is not a fraction and is deliberately unbounded: $\text{DLR} > 1$ indicates that static support moves more than the region intended to move, a regime that occurs in practice.

Drift Attenuation Ratio (DAR). DAR ($\downarrow$), one minus the ratio of compensated to raw dynamic-region flow, measures the signed relative attenuation of measured dynamic-region flow after global similarity compensation (Supp. S1). Headline tables report DAR clipped to $[0, 1]$ for compact comparison. Signed values are retained in the released records, and the incidence and magnitude of negative DAR are reported in Supplementary Table S14.

Interpretation. Neither is a causal decomposition of motion into drift and non-drift components. DLR and DAR are secondary diagnostics that divide the work between them: DLR is a translational-leakage diagnostic whose rotational response is inverted, so rotational drift is carried by fBD and DAR instead, and that division is why DLR is never read alone. The underlying DAR is signed, and negative values are retained. High DLR with moderate DAR indicates static-region motion a rigid model does not account for; it does not identify what the residual is.

Semantic guardrail. The decomposition is agnostic to what moves: a sequence rendering drifting fog where a river was requested can occupy a favorable operating point. The natural guardrail is a prompt-alignment score—CLIP-based image-text similarity [9], or a learned video-quality predictor [8] at higher cost. We do not report one, and the factors should be read accordingly: they establish where motion occurs and whether it persists, not that the depicted phenomenon is the one requested. Pairing SNF-Bench with a semantic score is how it should be used in practice.

4.4. Joint Interpretation

No factor is read alone: low drift with low compensated motion indicates an over-stabilized sequence, high compensated motion with high static-region flow indicates motion contaminated by displacement, and high retention with negligible magnitude indicates persistent but trivial motion. SNF-Bench therefore reports no single scalar score, since collapsing the factors would recreate the ambiguity it exists to expose.

Implementation convention. Videos are normalized to a common evaluation resolution and subsampled to a common flow-estimation rate; no frame interpolation is performed, since synthesizing frames would alter the flow field being measured. Every estimator, threshold, window extent and constant is fixed globally before the sweep and applied unchanged to every system, and the per-sequence frame rate, resolution and compensation mode are released alongside the scores.

5. Controlled Validation

Before SNF-Bench is used to compare generators, we validate that its measurements respond to the failure modes they target. A benchmark is not justified merely because its metrics rank differently; each must change selectively, monotonically, and in the expected direction under known corruptions. We therefore construct controlled variants of fixed-camera reference clips with known corruption type and severity, independently of the audit in Sec. 6: perturbation strengths, masks, windows, and hyperparameters are fixed before the final results are inspected.

5.1. Perturbations and Criteria

Given a reference sequence we generate variants that isolate individual components of the decomposition. Geometric drift applies a temporally increasing global transformation—translation, rotation, or scale—simulating camera-like background displace-

ment that accumulates as long-horizon error does. Motion attenuation progressively blends dynamic-region content toward an early reference, and late freezing replaces all content after a chosen fraction of the sequence, both leaving static support untouched. Photometric drift varies brightness and color temperature while preserving geometry and motion, bounding the residual sensitivity of flow-based quantities. Temporal repetition replaces late dynamic content with a short repeated cycle. Partition perturbation erodes and dilates the region boundary rather than the video.

Geometric severity is specified as the mean pixel displacement a corruption induces, with each family’s parameter derived from it using the frame geometry. This is not presentational: a parameter in degrees is not commensurate with one in pixels, and at the evaluation resolution one degree of rotation induces only a few pixels of displacement, so families compared by their native parameters differ in injected severity by more than an order of magnitude. A factor is admitted only if its response direction matches its definition, that response is monotonic in severity, unrelated corruptions produce substantially smaller changes, and conclusions survive perturbation of the region partition. Factors failing these checks are reported as diagnostics rather than headline quantities. The severity specification and the admission outcome for every factor are in the supplementary material.

Complementary checks. Mechanistic validation shows a factor responds to the corruption it names, not that its ordering matches what an observer perceives, and we do not close that gap. The supplementary material reports exploratory human and open-weight VLM pilots under an axis-specific forced-choice protocol. The human arm yielded too few decided pairs and low inter-rater agreement, while the VLM [26] passed the presentation-order control but failed the frozen-versus-moving control; neither arm supports a perceptual-validity claim.

5.2. Validation Results

Figure 2 and Table 1 report the outcome. Injected translation produces the expected monotone increase in both static-fidelity factors, and progressive late freezing drives MCFF-L and FP down—to 0.02 and 0.07 of their unperturbed values once the late window is frozen—while leaving the static factors comparatively unchanged, which is the selectivity the decomposition requires. At maximum translation, fBD and NBF reach $1.86\times$ and $1.32\times$ baseline, while VBench Dynamic Degree [12] reaches only $1.07\times$ and improves on a corruption fixed-camera evaluation should penalize.

Factor	target	ρ	ρ_{rot}	off-target	admitted
fBD	drift	+1.00	+0.70	14%	✓
NBF	drift	+1.00	+0.60	6%	✓
MCFF-L	freeze	+1.00	n/a	14%	✓
FP	freeze	+0.80	n/a	21%	✓
DLR	translational leakage	+1.00	-0.80	5%	✓
DAR	drift	+0.90	+1.00	18%	✓
VBench DD	drift	+0.70	+0.70	2%	—

Table 1. Mechanistic validation and factor admission. ρ is signed target-family Spearman correlation; ρ_{rot} reports rotational response; off-target is the largest relative excursion under photometric and mask perturbations. Admission requires $\rho \geq 0.7$ and off-target response below 25%. Rotation is reported but not gated: DLR is inverted there, while fBD and DAR carry that case. VBench DD is a contrast row, not a candidate factor. Twelve clips per severity.

Check	result
Static-mask area	0.614–0.663
Area–fBD Spearman	$\rho = -0.75$, exact $p = 0.066$, $n = 7$
No precipitation: NBF order	identical
No precipitation: fBD extremes	unchanged
Boundary perturbation: max. response	$\leq 14\%$

Table 2. Partition robustness at 60s. Static-region area is similar across checkpoints, and the headline NBF ordering is unchanged when precipitation is excluded. The negative area–fBD association has the direction expected under early-mask circularity and is therefore reported as an open validity concern rather than dismissed.

Two results qualify the framework rather than support it. Rotation is the harder geometric case for every factor, since correspondences degrade and the induced displacement is smallest at the frame center: both static-fidelity factors respond more weakly than under translation (+0.70 for fBD, +0.60 for NBF, against +1.00 for both), so neither is used alone. DLR is the sharper limitation—admitted on its target family, where it responds perfectly, but with an inverted rotational response (−0.80), so it is read only as a comparative diagnostic alongside DAR. Reported alongside these is the scope boundary of Sec. 5.1: looping content sustains both persistence and magnitude, so SNF-Bench cannot distinguish genuine progression from repetition, and does not claim to.

Table 2 summarizes the complementary partition checks; the full per-checkpoint sensitivity analysis is supplementary.

6. Public-Model Audit

We apply SNF-Bench to publicly released long-horizon generators. The goal is not a universal leaderboard but a determination of how checkpoints occupy the static-

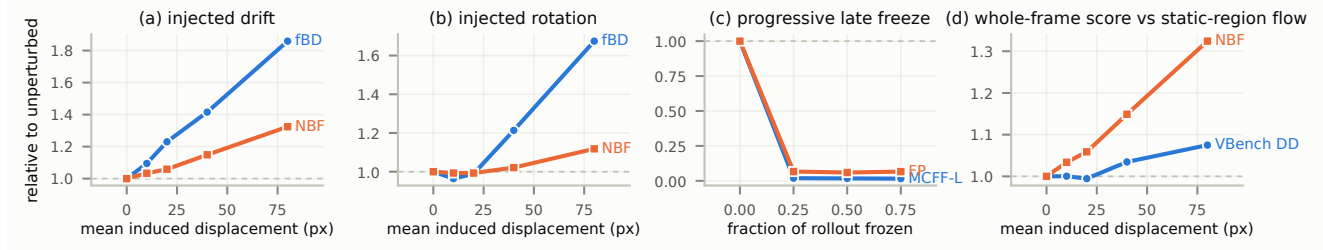


Figure 2. Mechanistic validation. Curves are relative to unperturbed fixed-camera clips. Translation raises fBD and NBF; progressive late freezing collapses MCFF-L and FP; under the same translation, VBench Dynamic Degree changes little and slightly upward. The perturbations are synthetic and implicate no published method.

515 fidelity/flow-persistence space under one recorded com-
516 mon T2V configuration.

517 Systems and setting. The audit covers 7 publicly
518 released autoregressive or streaming generators with
519 accessible checkpoints, frozen as a set before the
520 sweep. Every T2V checkpoint is loaded into the same
521 recorded inference implementation rather than its re-
522 leased pipeline. Fixed before the sweep to give the
523 distilled checkpoint family a shared step budget, the
524 configuration uses a four-step denoising schedule (in-
525 dices warped by the released scheduler), six frames
526 per block, seed 0, and one rollout/cache path. The
527 text-conditioned runs also share 832×480 output at 16
528 fps. Within a horizon, checkpoint weights and their
529 recorded regular/EMA selection are the only system-
530 specific inference inputs. Accordingly, these results
531 characterize checkpoints under this common configu-
532 ration, not each method’s published inference proce-
533 dure; a checkpoint distilled for another step budget
534 is not observed at its own operating point. The I2V
535 track instead separates released-pipeline outputs from
536 dagger-marked long-horizon-wrapper rows (Sec. 7.3).

537 Controls and statistics. Within each item every sys-
538 tem receives identical conditioning, and the seed is
539 fixed before the sweep. This makes each run re-
540 producible but does not make different checkpoints’
541 stochastic trajectories equivalent; cross-system com-
542 parison rests on the paired-item design. Videos are
543 retained in their recorded form and the normalization
544 of Sec. 4 applied unchanged. Scores are computed per
545 item and averaged over items; the sensitivity of ev-
546 ery conclusion to category macro-averaging is reported
547 in the supplementary material. Uncertainty is esti-
548 mated by percentile bootstrap [5] over items, and we
549 report 95% marginal intervals. These intervals sum-
550 marize per-checkpoint uncertainty; all headline sep-
551 aration claims use paired bootstrap distributions of

per-prompt differences, reported in Supplementary Ta-
ble S3. Measurement coverage and fBD abstentions
are reported in Supplementary Tables S6–S7; incom-
plete metric records are counted as partial rather than
silently treated as successful. To test whether the de-
composition changes interpretation, the same outputs
are additionally scored with established whole-frame
metrics and the induced orderings compared. The fo-
cus is interpretation change, not a claim that existing
metrics are wrong.

7. Results and Interpretation Changes

Unless stated otherwise, T2V comparisons refer to
checkpoints under the recorded common configuration.
I2V released-pipeline and wrapper results are identified
separately and never ranked across settings.

7.1. Recorded Common-Configuration T2V Audit

Table 3 reports the primary 60s T2V tier and a 120s
directional check. Because the 120s panel has 6 dif-
ferent prompts, it is not compared column-by-column
with 60s. The 5 and 240s diagnostics and all four
horizons are supplementary.

Motion persistence. We read $(MCFF(\mathcal{L}), FP)$ jointly:
sustained flow needs non-trivial late motion and high
retention. MCFF-E beside MCFF-L exposes uniformly
negligible motion.

Operating regimes. Joint static-fidelity and persis-
tence columns expose low-drift/low-motion, high-
motion/high-drift, and low-drift/non-trivial-motion
regimes without a scalar score. Infinite-Forcing [3]
has the lowest NBF under the recorded common
configuration—paired per-prompt intervals versus all
6 peers exclude zero (Supp. Table S3)—but also the
least surviving late motion. Causal-Forcing [43] occu-
pies the opposite, higher-background-flow corner.

Method	fBD↓	NBF↓	MCFF E→L	FP↑
60s horizon ($n = 23$)				
CausVid	8.08 ± 3.15	9.27 ± 4.14	2.87 → 1.96	0.67 ± 0.18
Self-Forcing	11.4 ± 3.63	12.7 ± 3.97	10.4 → 2.54	0.44 ± 0.22
Infinite-Forcing	7.75 ± 3.84	3.65 ± 0.88	1.17 → 0.29	0.63 ± 0.22
Rolling-Forcing	16.8 ± 3.69	11.2 ± 2.70	3.18 → 1.57	0.75 ± 0.29
Reward-Forcing	14.8 ± 3.71	8.51 ± 2.79	3.81 → 0.80	0.49 ± 0.18
LongLive	15.9 ± 2.89	13.9 ± 4.76	6.46 → 2.41	0.75 ± 0.28
Causal-Forcing	22.4 ± 2.26	87.6 ± 18.6	9.84 → 7.21	0.83 ± 0.31
120s horizon ($n = 6$)				
CausVid	6.15	3.42	0.36 → 0.52	1.06
Self-Forcing	21.2	6.80	4.85 → 0.82	0.85
Infinite-Forcing	16.1	2.60	0.50 → 0.32	0.64
Rolling-Forcing	23.8	10.6	3.33 → 0.77	0.38
Reward-Forcing	20.0	11.6	2.05 → 0.69	0.39
LongLive	22.5	9.66	5.10 → 1.19	0.81
Causal-Forcing	21.8	130	10.5 → 10.7	0.90

Table 3. Recorded common-configuration T2V audit. A four-step denoising schedule with released-scheduler warping, six frames/block, and seed 0. Header n is the maximum valid item count within each horizon; factor-specific abstentions are in Table S6. Units: fBD is % frame diagonal; NBF is 10^{-3} frame widths/s; MCFF is px/sampled interval; FP is unitless. FP is the mean of per-video clipped late/early ratios and therefore need not equal the ratio of the displayed aggregate MCFF-L and MCFF-E means. Means with percentile-bootstrap 95% CI half-width shown for compactness (10k resamples, seed 0); $n < 10$ shows means only. MCFF-E→L accompanies FP. Separation uses paired per-prompt intervals (Table S3). DLR/DAR are supplementary.

Method	fBD↓	NBF↓	MCFF E→L	FP↑
60s horizon ($n = 30$)				
Causal-Forcing++ (2-step) [†]	9.18	5.97	0.85 → 0.60	0.86
Causal-Forcing++ (1-step) [†]	5.34	7.05	0.86 → 0.57	0.77
Causal-Forcing (frame-wise) [†]	21.1	36.4	2.80 → 2.69	0.89
Causal-Forcing++ (2-step, frame-wise)	9.94	6.91	1.30 → 0.78	1.02
Self-Forcing [†]	17.3	5.52	1.50 → 0.69	0.89
CausVid [†]	19.9	3.31	0.58 → 0.41	0.81
Causal-Forcing++ (1-step, frame-wise)	10.4	5.25	0.53 → 0.52	1.01

Table 4. I2V released pipelines and deployment sensitivity. Unmarked rows are released-pipeline outputs; [†] rows use the wrapper; settings are not compared. Header n is the maximum valid item count within each horizon; factor-specific abstentions are in Table S6. Units: fBD is % frame diagonal; NBF is 10^{-3} frame widths/s; MCFF is px/sampled interval; FP is unitless. FP is the mean of per-video clipped late/early ratios and therefore need not equal the ratio of the displayed aggregate MCFF-L and MCFF-E means. Means; intervals are supplementary. MCFF-E→L accompanies FP. No cross-setting claim. DLR/DAR are supplementary.

7.2. Whole-Frame Motion and Static-Region Flow Yield Different Interpretations

Across 7 checkpoint means, Dynamic Degree tracks static-region flow at $\rho = +0.93$ (descriptive; per-item analysis in the supplementary material): more apparent motion also tends to mean more background motion. Causal-Forcing ranks 1/7 on Dynamic Degree but 7/7 on NBF; Infinite-Forcing mirrors it at 1/7 and 7/7. Causal-Forcing’s DLR of 0.94 indicates static-region motion energy nearly equal to dynamic-region energy. Its DAR of 0.40 means fitted similarity compensation reduces measured dynamic-region flow by that proportion; it is attenuation, not causal decomposition. The residual is unidentified and may include non-rigid instability, intended motion, partition contamination, or flow error.

7.3. Image-Conditioned (I2V) Track: Released Pipelines and Deployment Sensitivity

At 60s, Table 4 separates two released-pipeline outputs from five dagger-marked wrapper rows; it is not a seven-system audit, and settings are never ranked. The panel shows deployment sensitivity without attributing its cause. Unlike T2V, each system shares one source image, so static support is given rather than imagined differently. Complete horizons and 5s bidirectional references are supplementary.

8. Limitations and Conclusion

SNF-Bench is limited to long-horizon fixed-camera generation where static support and dynamic flow separate. It measures neither physical nor semantic realism, cannot distinguish progression from repetition, and depends on an automatic partition bounded by our erosion-dilation study.

Whole-frame motion and region-resolved static fidelity can materially change the interpretation of the same outputs. Reporting static fidelity and flow persistence separately, with drift quantities only as context, complements broad benchmarks without a composite score. The released specification and common-configuration checkpoint audit provide a reproducible basis for evaluation; independent partitions and broader non-water coverage remain important next steps.

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